frontline education™
Absence Management

INTEGRATE ABSENCE MANAGEMENT WITH YOUR OTHER SYSTEMS VIA WEB SERVICE:

PROJECT OVERVIEW

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Overview

Absence Management is a Software as a Service (SaaS) application from Frontline Education, and it provides a complete absence management and substitute placement management service for the K-12 Education market.

Absence Management has been developed with integrations in mind. It is critical that applications like this, which track employee leave and time worked (both employee and substitute), are able to **communicate with** other applications in a district office such as **Payroll, Time & Attendance, Human Resources, Applicant Tracking, Professional Development, and others**.

The integration abilities of Absence Management are outlined below, in their two general categories – Import and Export.

Import – Demographic Records

One of the most common integration points connecting into Absence Management is that of passing in all fields necessary to setup, modify or terminate employee and substitute demographic records (Add/Update/Delete). There are several methods in Absence Management to allow for this modification of data.

1. Manual File Import – Absence Management offers simple templates for many data fields in the Absence Management system for employees and substitutes. These templates are housed directly in the Absence Management site as well as the import utility itself. This allows district users to easily push data in to Absence Management for many functions.
    a. Each Import Template comes with its own Users Guide showing all the field uses, lengths and any flexibility of formatting – as well as step by step instructions for importing through the built-in import utility
    b. These templates are easy to use and give districts flexibility in managing data at any time. Import file formats are acceptable in either XLS or CSV.
    c. Fields, FAQs and details in Appendix A
2. Automatic File Import – Absence Management also offers an automatic option for importing batch files.
    a. Files of data are automated by the district processes to a secure FTP folder (SFTP, FTPS or even FTP) which is hosted by Frontline Education. Automatic jobs pick up and auto-load the files on a scheduled basis with no further intervention needed by the district. Email alerts are generated and sent for each file operation which notifies the district contact of the success of the import and any errors. A detailed error report is also made available within the application.
3. Web Services – Absence Management offers a more real time communication methods through web services.

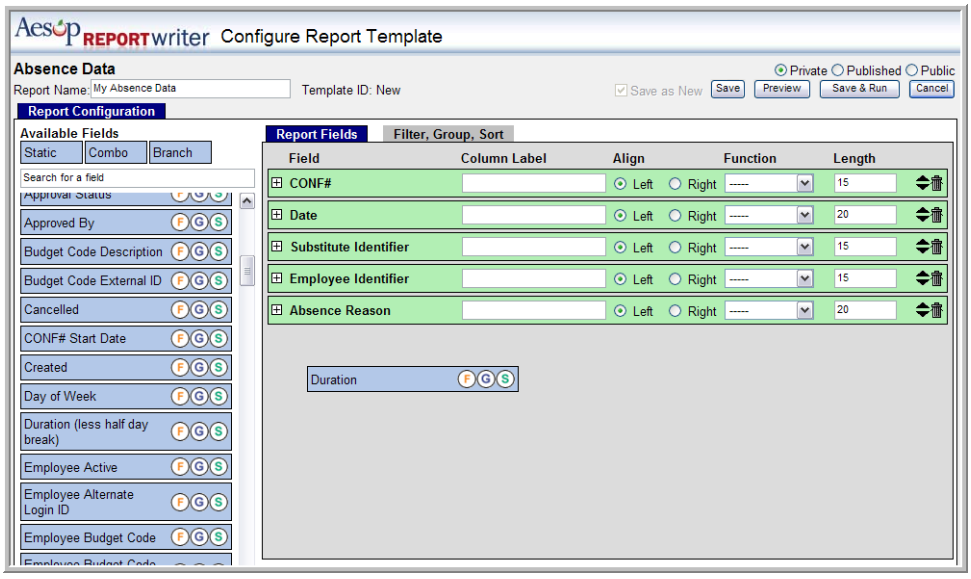
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a. The district can send XML packets of information in a real time transaction which is consumed by Absence Management for both **Adds** and **Updates** to demographic and skill fields.
b. This method provides a more transaction-based and near real time approach as opposed to loading flat files on a scheduled basis.
c. Technical specifications:
    i. Web Service Address http://www.aesoponline.com/webservices/dataimport.asmx
    ii. Use AddOrUpdate method to get normal behavior. Add method will not update an existing record.
    iii. A negative return code in the result XML denotes an error.
    iv. Syntax and FAQs in Appendix B

Export – All Data

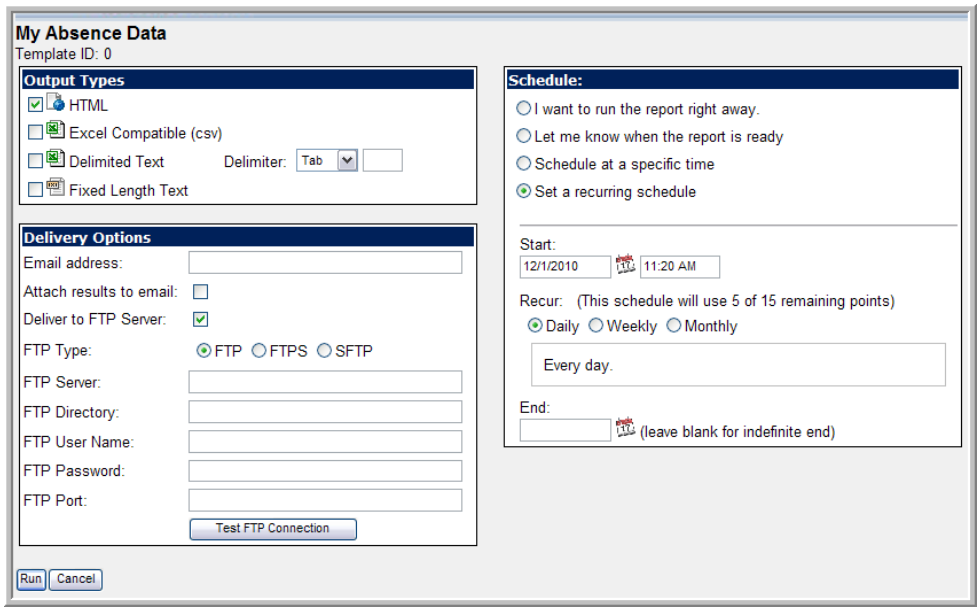
Absence Management contains one of the most flexible and easy-to-use export tools in the K12 Education market called the Absence Management Report Writer. Below are the two main export methods for Absence Management integrations.

1. Report Writer custom extract designer – this feature is built directly into the Absence Management application (proprietary design – not a third-party reporting tool) and allows districts to design custom extract files of any field in the Absence Management system.
    a. The Report Writer utilizes a simple drag-and-drop function to allow districts the creation of perfect extract files.



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b. Over 300 data fields are available including the ability to specify rounding rules, if/then/else statements, custom logic and field formatting.
c. Filtering, Grouping and Sorting is also a native feature.
d. Export options include Immediate, Recurring, and more.



e. Files can be scheduled to any FTP, SFTP or FTPS site or even an email address.
f. The main benefit of the Report Writer is its ability to act as a complete custom report and extract designer.

2. Web Services – much like the Import method, many districts and vendors prefer to receive employee absence and sub work data via web service.
    a. Absence Management's Report Writer acts as the gateway for our export API. Districts can place calls to the web service and query reports (Template IDs) that are custom designed by them. These reports contain the defined fields and formatting that tells Absence Management what data elements to return via the web service.
    b. The more real time communication gives districts the same access to Absence Management's database as if Absence Management were an on-premise solution.
    c. Technical Specifications are available in Appendix C

Frontline believes that in addition to all the available integration technology within the Absence Management system, expert personnel are also a key part of any integration project. Frontline's Technical Services & Integrations Team is standing by to assist with any question or process. The Technical Services & Integrations Team covers a wide variety of specialties including development, customer service and best practices advising. The team is available to both our district clients and our partner vendors.

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Appendix A.1 – File Import Specifications, Employee File (Employee Demographics, Employee Skills, Absence Balances)

This guide will walk through the available fields in the **Employee Import Template** and what they mean. The Import Template is broken down in highlighted sections based on what type of information you are trying to import. They are broken down as follows…

Blue – Employee general info and permissions
Highlighted Blue – Required fields for importing new employees
Orange – Preferred substitute information
Pink – Skill information
Green – Absence balance information

Notes:
If creating new records, all **Highlighted Blue** fields are required.

If updating existing records, only the **Identifier** and the fields being updated are required.

Every employee import will require the **Identifier** column so Absence Management knows which employee to make the changes to.

Excel Compatible (csv) Column Header	xml	Description
Identifier	Identifier=""	District unique Id number. May be alphanumeric and up to 30 characters . **Recommend an Employee number that matches the employee's ID from your payroll/hr system. **This field will be required in all data imports to identify the employee you want to update**
Last Name	LastName=""	Employee's last name. Up to 30 characters
First Name	FirstName=""	Employee's first name. Up to 30 characters
Middle Name	MiddleName=""	Employee's middle name. Up to 10 characters
Title	Title=""	Employee's specific title like "Second Grade" or "Nurse"
Phone	Phone=""	Employee's phone number. This will also be the login id. Do not use spaces or dashes Example: 5556867848
Alternate Login ID	AlternateLoginID=""	This is used as an alternate option to using the phone number as the Login ID. (8,9,11-20 characters)
Pin	Pin=""	This will be the employee's PIN to login. Leave blank for new employees if you want

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		Absence Management to auto-assign a PIN. Must be 4 or 5 digits and only numeric
School External ID	SchoolExternalId=""	The external ID of the school you want to assign the employee to. The external ID is a district assigned ID and can be found on the school's general information page. Don't use this field if you are using "Inst_ID" in your import
Inst_ID		The Web ID of the school you want to assign the employee to. The Web ID can be found on the school's general info page. Don't use this field if you are using "School External ID" in your import
Employee Type Description	EmployeeTypeDescription=""	The employee type you want to assign to the employee. Must match an employee type in Reference Data
Employee Type ID	EmployeeTypeId=""	The ID associated with the Employee Type Description. The Employee Type ID can be found in the Reference Data
Substitute Requirements	NeedsSubstitute=""	Does this employee require a sub? **O** or **2**= Employee can decide if sub is required **N** or **0** = Never requires a sub **Y** or **1** = Always requires a sub If you leave this field blank for the Excel Import for new employees it will default to **Y**
Room	Room=""	Put "Main Office" in this field
Email	Email=""	Employee's email address Example: employee@school.edu
Street 1	Street1=""	Employee's Street Address Example: 501 Trooper Rd
Street 2	Street2=""	Address part two Example: Apt 1
City	City=""	The city in the employee's address
State	State=""	State code for the address Example: PA or OH
Zip	Zip=""	Zip Code for the address Example: 19464 or 19464-1245
External ID	ExternalID=""	Miscellaneous data field on the employee's general info page
External ID 2	ExternalID2=""	Miscellaneous data field on the employee's general info page
External ID 3	ExternalID3=""	Miscellaneous data field on the employee's general info page
Start Time	StartTime=""	Employee's start time Example: 7:00 AM or 07:00
End Time	EndTime=""	Employee's end time

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		Example: 3:00 PM or 15:00
Half Day Break	HalfDayBreak1=""	This would be the end time of a half day am absence Example: 11:00 AM or 11:00
Half Day Break 2	HalfDayBreak2=""	This would be the start time of a half day pm absence Example: 11:00 AM or 11:00
Remove Employee	Deleted=""	Put a **Y** in this column to delete the employee from Absence Management
Active	Active=""	Put a **Y** in this column to mark the employee as active Put a **N** in this column to mark the employee as inactive Leaving this column blank for new employees will default to **Y**
Can Assign Substitute	CanAssignSub=""	Put a **Y** in this column to allow the employee to assign substitutes Put a **N** in this column to not allow the employee to assign
Can Cancel Absence	CanCancel=""	Put a **0** (zero) in this column to not allow the employee to cancel absences Put a **1** in this column to allow the employee to cancel only unfilled absences Put a **2** in this column to allow the employee to cancel both filled and unfilled absences
Can Upload Files	CanUpload=""	Put a **Y** in this column to allow the employee to upload files to absences Put a **N** in this column to not allow the employee to upload files to absences
Notes	Notes=""	Notes about the employee that will appear on their general info page. Character limit is 50
Accounting Code External ID	AccountingCodeExternalID=""	External ID of the accounting code you want to assign to the employee
Budget Code External ID	BudgetCodeExternalID=""	External ID of the budget code you want to assign to the employee
Birth Date		Employee's birth date Example: 5/25/1977 or 05/21/1980
Start Date		Employee's employment start date Example: 9/25/2002 or 11/5/2008
End Date		Employee's employment end date Example: 2/10/2009 or 07/15/2007
Preferred Substitutes List Permission		Allow the employee to edit their preferred sub list **0** = Use default setting at school level **1** = Do not display list **2** = Display list only **3** = Edit list (basic)

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		4 = Edit list (adv.) **6** = Edit List (Basic) – No Exclude **7** = Edit List (Adv.) – No Exclude
Hours Per Day	HoursPerDay=""	Enter the number of hours the employee works per day Example: **8** for eight hours
Email Approval Updates		Do you want this employee to receive email updates of approval status? **Y** for Yes **N** for No
Language Description		Default IVR language the employee hears on the phone. **English** **Spanish** **French**
Gender		**M** for Male **F** for Female
Ethnicity Description		Must match the an ethnicity description in reference data
Skill Code		The Skill Code from reference data of the skill you want to assign to the employee
Skill Type		What type of skill? **MH** for Must Have **SH** for Should Have **MNH** for Must Not Have
Remove Skill		Put a **Y** in this column to remove the skill
Absence Reason External ID		The external ID from reference data of the absence reason you want to give the employee a balance for
Balance		The balance in days or hours (depending on the employee's settings) of the absence reason Example: **3** for 3 days or hours
As Of Date		The date you want the balance to begin being calculated Example: **9/1/2010** or **09/01/2010**
Remove Absence Reason		Put a Y in this column to remove the absence reason
Substitute Identifier		The Identifier of the Substitute you want to add to the employee's preference list
Remove Substitute		To remove the substitute from the employee's preference list put a **Y** in the column
Exclude		To mark the sub as excluded from the employee's preference list put a **Y** in this column
Visibility Percentage		Set the override visibility percentage for the substitute

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		Example: put **90** for 90 percent
Favorite Rank		Set the favorite five rank for this substitute. You can **choose from 1 to 5**

Appendix A.2 – File Import Specifications, Substitute File (Substitute Demographics, Substitute Skills)

This guide will walk through the available fields in the **Substitute Import Template** and what they mean. The Import Template is broken down in highlighted sections based on what type of information you are trying to import. They are broken down as follows…

Blue – Substitute general info and permissions
Highlighted Blue – Required fields for importing new substitutes
Pink – Skill information
Orange – Preferred school information

Notes:
If creating new records, all **Highlighted Blue** fields are required.

If updating existing records, only the **Identifier** and the fields being updated are required.

Every substitute import will require the **Identifier** column so Absence Management knows which substitute to make the changes to.

Excel Compatible (csv) Column Header	xml	Description
Identifier	Identifier=""	District unique Id number. May be alphanumeric and up to 30 characters. The substitute Identifier typically matches the Sub's ID from the district's Payroll/Hr system. **This field will be required in all data imports to identify the substitute you want to update**
Pin	Pin=""	This will be the substitute's PIN to login. Leave blank for new substitutes if you want Absence Management to auto-assign a PIN. Must be 4 or 5 digits and only numeric
Last Name	LastName=""	Substitute's last name. Up to 30 characters
First Name	FirstName=""	Substitute's first name. Up to 30 characters
Middle Name	MiddleName=""	Substitute's middle name. Up to 10 characters
Phone	Phone=""	Substitute's phone number including the area code. This will also be their login id. Don't use spaces or dashes Example: **5556867848**

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Email	Email=""	Substitute's email address Example: **substitute@school.edu**
Active	Active=""	Put a **Y** in this column to mark the substitute as active Put a **N** in this column to mark the substitute as inactive Leaving this column blank for new substitute will default to **Y**
Remove Substitute	Deleted=""	Put a **Y** in this column to delete the substitute from Absence Management
Can Search On Web	CanSearchForJobsOnWeb=""	Put a **Y** in this column to allow the sub to search for jobs Put a **N** in this column to not allow a sub to search for jobs Leaving this column blank for a new sub will default to **Y**
Can Cancel	CanCancel=""	Put a Y in this column to allow the sub to cancel jobs Put an N in this column to not allow the sub to cancel jobs Leaving this column blank for a new sub will default to **N**
Permanent Sub School DbKey		The **Web ID** of the school that you want to assign the sub as a permanent sub. The Web ID can be found on the school's general info page
Permanent Sub Rank		The permanent sub rank you want to give the substitute **1** for First **2** for Second Ect…
Pay Code External ID	PayCodeExternalID=""	The **external ID** of the pay code you want to assign to the sub. Must match the pay code external id found in reference data
Notes	Notes=""	Notes about the substitute that will appear on their general info page. Character limit is 50
Gender		**M** for Male **F** for Female
Language Description		Default IVR language the substitute hears on the phone. **English** **Spanish** **French**
External ID	ExternalID=""	Miscellaneous data field on the substitute's general info page
Send Email Notifications	SendEmailNotifications=""	Do you want the sub to receive email notifications from Absence Management? **Y** for Yes

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		N for No
Preferred School List Permission		Set the permission for the substitutes preferred school list **Hide** = Do not display list **Display** = Display List **Edit** = Edit List
Date of Join		Start date of the substitute Example: **9/15/2008** or **09/09/2009**
Date of Birth		Substitute's birth date Example: **5/25/1977** or **05/21/1980**
External ID 2		Miscellaneous data field on the substitute's general info page
External ID 3		Miscellaneous data field on the substitute's general info page
Street 1	Street1=""	Substitute's Street Address Example: **501 Trooper Rd**
Street 2	Street2=""	Address part two Example: **Apt 1**
City	City=""	The city in the substitute's address
State	State=""	State code for the address Example: **PA** or **OH**
Zip	Zip=""	Zip Code for the address Example: **19464** or **19464-1245**
Skill Code		The Skill Code from reference data of the skill you want to assign to the substitute
Remove Skill		Put a **Y** in this column to remove the skill
Skill Expiration		The date the skill expires for the substitute Example: **8/31/2011** or **08/31/2011**
Preferred School List Type		Does the sub want to see jobs at the school selected or not want to see jobs at the school selected? **Include** = wants to see jobs **Exclude** = does not want to see jobs
Preferred School DbKey		The **Web ID** of the school you would like to add to the sub's preferred school list. The school's Web ID can be found on the school's general info page
Remove Preferred School		Put a **Y** in this column to remove the school from the sub's preferred school list

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Appendix B – Import Web Service Specifications

Employee Syntax: (blue indicates a required field for a new record)

<employees>
<!—employee record -->
<employee
DbKey=""
Identifier=""
NewIdentifier=""
SchoolExternalId=""
RemoveFromThisSchool=""
RemoveFromAllOtherSchools=""
FirstName=""
MiddleName=""
LastName=""
Title=""
AlternateLoginID=""
EmployeeTypeId =""
EmployeeTypeDescription=""
Pin=""
Phone=""
Email=""
Room=""
Street1=""
Street2=""
City=""
State=""
Zip=""
StartTime=""
HalfDayBreak1=""
HalfDayBreak2=""
EndTime=""
HoursPerDay=""

ExternalID=""
ExternalID2=""
ExternalID3=""
ExternalID4=""
ExternalID5=""
ExternalID6=""
ExternalID7=""
ExternalID8=""
ExternalID9=""
ExternalID10=""

BudgetCodeExternalID=""

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AccountingCodeExternalID=""

NeedsSubstitute=""
CanAssignSub=""
CanCancel=""
CanUpload=""
CanHoldAbsenceProcessing=""
CanHoldAbsenceTill=""
CanViewSubReportTimes=""
CanEditSubReportTimes=""
TimesToCallPrefSubList=""
EditPreferredSubList=""
EmailApprovalStatus=""
Notes=""
Active=""
Deleted=""
/>

</employees>

<!—Alternate syntax for multiple schools (all other employee fields allowed too) -->
<employees>
<employee DbKey="" Identifier="" RemoveFromAllOtherSchools="">
<school
SchoolWebId="" SchoolExternalId=""
RemoveFromThisSchool=""
/>
<school
SchoolWebId="" SchoolExternalId=""
RemoveFromThisSchool=""
/>
</employee>
</employees>

Substitute Syntax: (blue indicates a required field for new records)

<substitutes>
<substitute
DbKey="" Identifier=""
NewIdentifier=""
FirstName=""
MiddleName=""
LastName=""
AlternateLoginId=""

Pin=""
Phone=""
Email=""

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```
Street1=""
Street2=""
City=""
State=""
Zip=""

ExternalId=""
PaycodeExternalId=""
CanSearchForJobsOnWeb=""
CanCancel=""
SendEmailNotifications=""
Notes=""

Active=""
Deleted=""
/>
</substitutes>
```

Sample Employee and Substitute Packets:

```
<?xml version="1.0" encoding="utf-8"?>
<soap12:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soap12="http://www.w3.org/2003/05/soap-envelope">
  <soap12:Body>
    <AddOrUpdate xmlns="http://tempuri.org/">
      <Data>
        <substitutes>
          <substitute dts="9/21/2009 5:56:45 PM"
GUID="866002C6F1A54E37A5AAD261F4513DFC" Action="34"
           Identifier="1234" ExternalID="Stamp-9/21/2009 5:56:45 PM"
                       FirstName="Stephanie" MiddleName="L" LastName="Doe"
                       Phone="8005559999" Street1="9000 Main Street" City="Santa
Elvira" State="CA" Zip="9000"
                       Gender="F" BirthDate="08/26/1983"
EmploymentStartDate="12/11/2006" Deleted="" Active="1"
                                                Pin="1234" title="SUBSTITUTE" />
        </substitutes>
      </Data>
      <WebServiceKey>XXXXXXXXX</WebServiceKey>
    </AddOrUpdate>
  </soap12:Body>
</soap12:Envelope>

<?xml version="1.0" encoding="utf-8"?>
<soap12:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soap12="http://www.w3.org/2003/05/soap-envelope">
  <soap12:Body>
```

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```
    <AddOrUpdate xmlns="http://tempuri.org/">
      <Data>
        <employees>
          <employee dts="2/16/2009 09:00 AM"
GUID="7D0909B0C3FB49C09271DA451DC1C164" Action="34"
Identifier="111111"
ExternalID="STAMP-2/16/2009 09:00 AM"
Pin="1234"
FirstName="Carl"
MiddleName="Ullysses"
LastName="Frederickson"
Phone="8885551212"
Street1="666 The Beast Drive"
City="San Hombre"
State="CA" Zip="99999"
Gender="M"
BirthDate="01/15/1949"
EmploymentStartDate="08/01/1990"
Active="1"
Room="Main Office"
MsgRequired="1"
EmailApprovalStatus="1"
RemoveFromAllOtherSchools="1"
EmployeeTypeDescription="Teacher"
NeedsSubstitute="2"
CanCancel="1"
CanUpload="1"
EditPreferredSubList="1"
HomeCampusExternalId="666"
SchoolExternalId="666"
Title="Teacher, Elementary" /></employees></Data>
<WebServiceKey>XXXXXXXXX</WebServiceKey>
</AddOrUpdate>
</soap12:Body>
</soap12:Envelope>
```

Sample Employee and Substitute Packets, cont:

File used to add and remove schools from existing employee:

```
<?xml version="1.0" encoding="utf-8"?>
<employees>
  <employee Identifier="70" Active="1">
    <school SchoolExternalId="030 " RemoveFromThisSchool="0" />
    <school SchoolExternalId="180 " RemoveFromThisSchool="0" />
    <school SchoolExternalId="165" RemoveFromThisSchool="1" />
  </employee>
</employees>
```

File used to add a De-Activate existing sub:

```
<?xml version="1.0" encoding="utf-8"?>
```

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```
<substitutes>
  <substitute Identifier="5648" Active="0" />
</substitutes>
```

File used to update existing sub:

```
<?xml version="1.0" encoding="utf-8"?>
<substitutes>
  <substitute Identifier="6870" FirstName="SCARLETT" MiddleName="A"
LastName="BROWN" Street1="8409 THIRD STREET" Street2="" City="ANYTOWN" State="KY"
Zip="40000" Active="1" />
</substitutes>
```

File used to add a new employee:

```
<?xml version="1.0" encoding="utf-8"?>
<employees>
  <employee Identifier="1450" Deleted="0" SchoolExternalId="001 " Room="Main
Office" FirstName="GREGORY" LastName="JACKSON" Title="SCHOOL DISTRICT DEPUTY
SUPT" EmployeeTypeDescription="Superintendent" ExternalID10="0020" Pin="31450"
Phone="7077073019" Email="GREGORY.JACKSON@psst.com" Street1="9054 EIGHTEENTH
STREET" Street2="" City="ANYTOWN" State="KY" Zip="40000" StartTime="1/1/1900
12:00:00 AM" HalfDayBreak1="1/1/1900 12:00:00 AM" HalfDayBreak2="1/1/1900
12:00:00 AM" EndTime="1/1/1900 12:00:00 AM" NeedsSubstitute="0" Active="1">
    <school SchoolExternalId="001 " RemoveFromThisSchool="0" />
  </employee>
</employees>
```

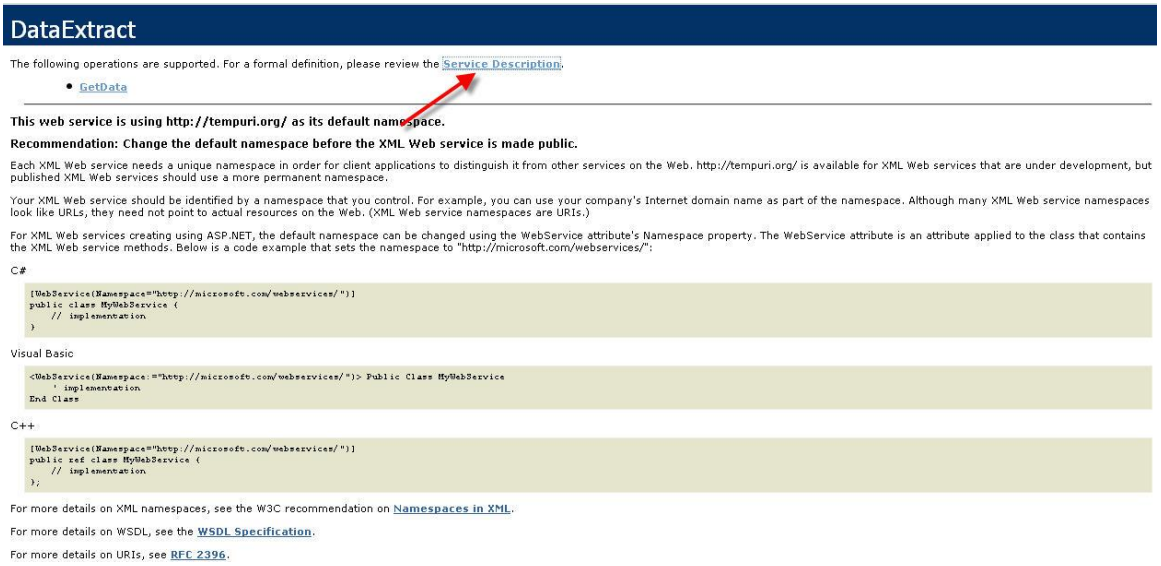
FAQ:

1. For Add records, all required fields must be sent
2. For Update records, only the fields needing to be updated should be sent
3. Testing can be accomplished using the test harness at the source website
4. A unique API Key is attached to each district and is required for web service access

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Appendix C – Export Web Services through Report Writer

Open a new tab in Internet Explorer and browse to http://www.aesoponline.com/webservices/dataextract.asmx to see this…



Click the link SERVICE DESCRIPTION to obtain the WSDL for the service. If the WEB SERVICE API Key is needed, then please obtain from the Frontline Technical Services & Integrations Team.

And when you need the report ID, then you have the ID from Step 1. You can also click the GETDATA link on the page from Step 2 for testing the format and data returned.

The report ID is from a report which is custom built to export the exact data elements you require, upon the web service call.

Example of Filter String in GetData:

```
<Filters mode="add">
<Filter name="Date" value1="9-15-2008" value2="9-30-2008" />
</Filters>
```

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